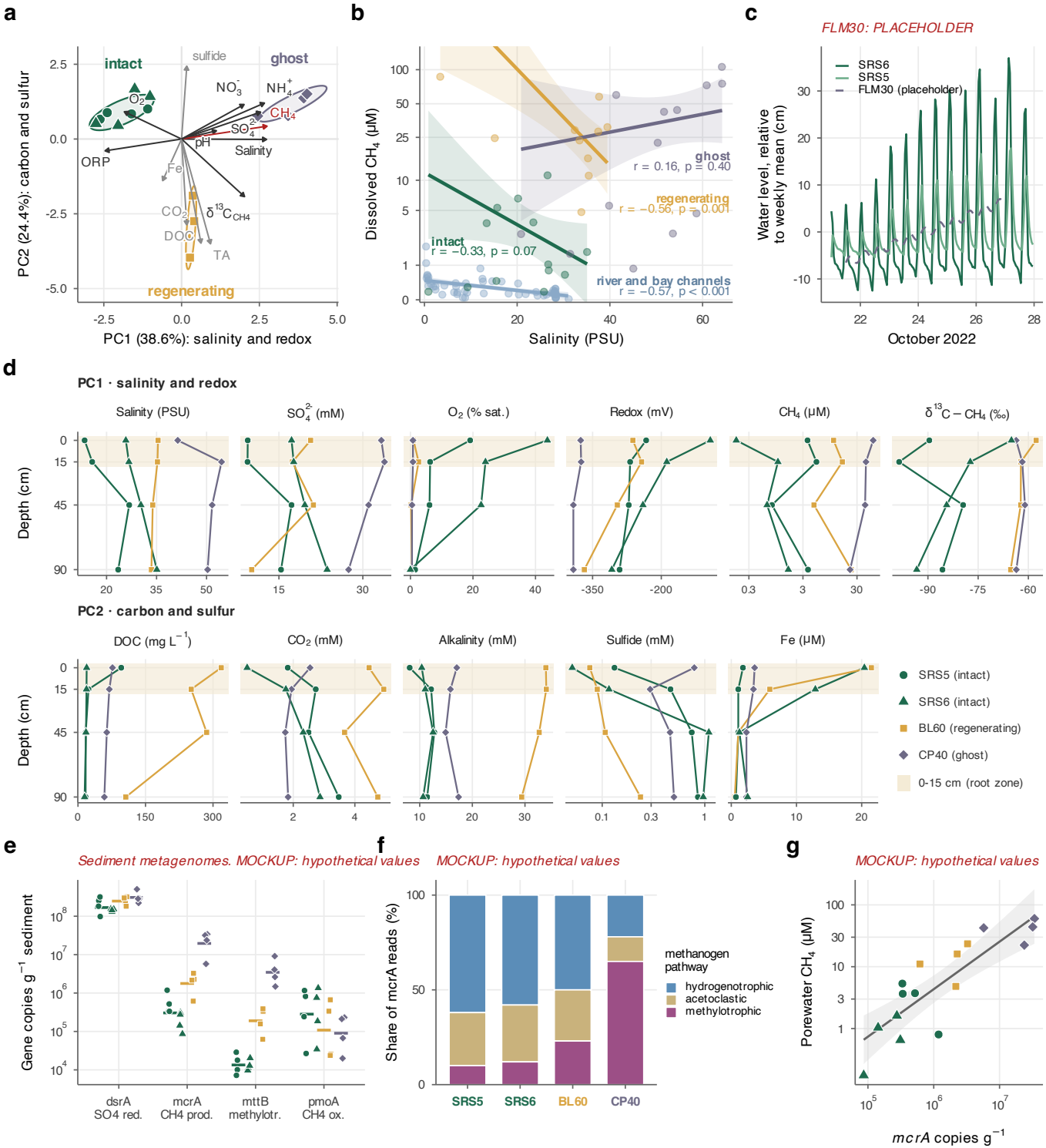

a
intact
ghost
sulfide
NO_3^-
NH_4^+
O_2
CH_4
SO_4^{2-}
pH
Salinity
ORP
Fe
CO_2
$\delta^{13}C_{CH4}$
DOC
TA
regenerating
PC2 (24.4%): carbon and sulfur
PC1 (38.6%): salinity and redox
b
Dissolved CH_4 (µM)
ghost
r = 0.16, p = 0.40
regenerating
r = −0.56, p = 0.001
intact
r = −0.33, p = 0.07
river and bay channels
r = −0.57, p < 0.001
Salinity (PSU)
c
FLM30: PLACEHOLDER
SRS6
SRS5
FLM30 (placeholder)
Water level, relative to weekly mean (cm)
October 2022
d
PC1 · salinity and redox
Salinity (PSU)
SO_4^{2-} (mM)
O_2 (% sat.)
Redox (mV)
CH_4 (µM)
$\delta^{13}C - CH_4$ (‰)
Depth (cm)
PC2 · carbon and sulfur
DOC (mg L^{-1})
CO_2 (mM)
Alkalinity (mM)
Sulfide (mM)
Fe (µM)
SRS5 (intact)
SRS6 (intact)
BL60 (regenerating)
CP40 (ghost)
0-15 cm (root zone)
e
Sediment metagenomes. MOCKUP: hypothetical values
Gene copies g^{-1} sediment
dsrA
SO4 red.
mcrA
CH4 prod.
mttB
methylotr.
pmoA
CH4 ox.
f
MOCKUP: hypothetical values
Share of mcrA reads (%)
methanogen pathway
hydrogenotrophic
acetoclastic
methylotrophic
SRS5
SRS6
BL60
CP40
g
MOCKUP: hypothetical values
Porewater CH_4 (µM)
mcrA copies g^{-1}